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Origami Oscillator R&D: Synthesis Modes Beyond the Wavetable

Executive summary

Origami has an opportunity to differentiate itself by treating an oscillator mode as a **different generative model of sound**, not as another collection of wavetable warps. That distinction matters because the current competitive field is already extremely broad: Serum 2 now combines wavetable, sample, multisample, granular, and spectral oscillators; Pigments spans wavetable, granular, modal, additive/harmonic, sample, and virtual analog; Phase Plant combines wavetable, granular, sample, analog/noise generation and extensive audio-rate modulation; and Vital pushes wavetable synthesis deeply into harmonic-domain spectral warping. [1] The strongest whitespace is therefore **stateful, spatial, rule-based, event-based, and emergent oscillation**: modes whose internal object is a field, evolving geometry, physical network, population, rule system, or previous cycle rather than a waveform or list of partials. I recommend prototyping **FIELD**, **GENDRIFT**, and **QUASAR** first: FIELD is the brand-defining visual/sonic concept; GENDRIFT is unusually novel yet straightforward to make pitch-stable; QUASAR gives immediate musical utility and aggressive bass/formant sounds at comparatively modest CPU cost. After those, KINETIC and AUTOMATA are the strongest candidates for a second wave. The overarching engineering principle should be: **every mode gets one musically stable phase clock, deterministic state when desired, native stereo behavior, and an explicit anti-aliasing policy** rather than relying on brute-force oversampling alone. Vital's emphasis on Nyquist-limited oscillator output and efficient unison, Phase Plant's explicit pre-modulation bandlimiting, and recent research on finite Fourier representations all reinforce that alias control is a first-class oscillator-design problem rather than an afterthought. [2]

Competitive whitespace and design rules

A useful test for every Origami oscillator concept is:

> **If the central idea can be reduced to “take a wavetable and warp/filter/shift/FM it,” it does not deserve to be an oscillator mode.**

Serum 2 already allows filtering, FM, phase distortion, ring modulation, distortion and dual warps around its wavetable oscillator, while its spectral oscillator resynthesizes samples at the harmonic level. [3] Vital explicitly describes its core differentiation as spectral warping that manipulates a waveform's harmonics. [4] Phase Plant exposes phase, frequency and amplitude modulation at audio rate and already provides a wavetable bandlimit stage specifically for material that will undergo heavy phase modulation. [5] In other words, **“new harmonic transformation” is crowded territory**.

Likewise, plain modal, additive, virtual analog, granular and sample-based generation are not strong headline differentiators for Origami because Pigments explicitly ships all six of those synthesis families, and Serum 2 now covers several of the same categories. [6]

Existing platform	Oscillator/generator territory already covered	Implication for Origami
Serum 2	Wavetable, multisample, sample, granular, spectral; FM/PD/ring/distortion/dual warps	Do not make another warp-heavy oscillator and call it new. [3]
Vital	Wavetable + harmonic/spectral warping, audio-rate modulation, stereo modulation	A “spectral tension/shift/stretch” engine has weak differentiation. [7]
Phase Plant	Analog, noise, sample, granular, wavetable; audio-rate phase/frequency/amplitude modulation	Plain FM/PM or modular oscillator routing is already expected. [5]
Pigments	Modal, granular, wavetable, sample, harmonic/additive, virtual analog	Plain physical modal/additive/VA engines are no longer exotic categories. [8]

The proposed Origami architecture should instead give every mode a genuinely different **internal state object**:

Origami mode	Thing the user is really manipulating
Wavetable	Shape / frames
Granular	Fragments of time
Spectral	Frequency components
FIELD	Space
GENDRIFT	Evolving geometry
QUASAR	Pitch-synchronous events
KINETIC	Coupled physical state
AUTOMATA	Rules and generations
HIVE	A self-organizing population
RECURSION	Memory of previous cycles
COLLIDER	Interactions and impacts
CIPHER	Discrete computation
DAEMON	Nonlinear dynamical state

“Novel” below means **uncommon and differentiating as a mainstream wavetable-synth workflow relative to the products surveyed**, not a claim of academic or patent novelty. Several modes deliberately turn established DSP or computer-music primitives into a different, musically constrained instrument design. Scanned synthesis, for example, already uses a mass/spring network plus a trajectory through that network, and dynamic stochastic synthesis already evolves waveform control points probabilistically; those are valuable precedents precisely because Origami can turn them into much more accessible, pitch-disciplined oscillator experiences. [9]

Prioritized oscillator concepts

Concept matrix

Scores are product-R&D judgments on a 1–5 scale. **Feasibility** includes not merely getting sound, but getting repeatable pitch, modulation stability, useful presets and a UI that musicians can understand.

Priority	Mode	Core representation	Novelty	Feasibility	User value	CPU
★	**FIELD**	2D scalar/vector environment	5.0	4.0	5.0	Med–High
★	**GENDRIFT**	Evolving cycle geometry	5.0	5.0	5.0	Low–Med
★	**QUASAR**	Analytic event packets	4.5	5.0	5.0	Low–Med
4	**KINETIC**	Coupled mass/spring state	4.5	4.0	4.5	Medium
5	**AUTOMATA**	Cellular/rule state	5.0	4.5	4.0	Low
6	**HIVE**	Coupled oscillator population	5.0	3.5	4.5	Med–High
7	**RECURSION**	Previous-cycle memory	5.0	3.5	4.5	Medium
8	**CIPHER**	Bitwise/state-machine rules	4.5	5.0	4.0	Low
9	**COLLIDER**	Phase-locked particle impacts	5.0	3.0	4.0	Medium
10	**DAEMON**	Nonlinear state-space machine	5.0	2.5	4.0	High

FIELD — “The oscillator is a world, not a waveform.”

Tagline: Move through space. Hear the terrain.

FIELD contains a two-dimensional scalar or vector environment $F(x, y)$, built from attractors, repellers, vortices, ridges, procedural texture and other spatial primitives. A MIDI-pitch-locked trajectory $P(\phi)$ passes through that environment, producing audio as $s(\phi) = F(P(\phi))$; left and right can use spatially separated trajectories instead of duplicated/detuned voices.

The crucial difference from wavetable synthesis is that you are not editing a one-dimensional waveform or scanning frames. You alter the **environment that produces the waveform**, so moving a node, rotating the trajectory or disturbing the flow changes the sound according to spatial relationships.

Problems FIELD directly attacks: static frame-by-frame timbral motion; stereo width that depends on detuned unison; no native concept of spatial timbre; reliance on prebuilt wavetable content; abstract parameter names whose effect is hard to visualize; modulation that feels like parameter automation instead of manipulating the sound-producing object. Modern synth interfaces already demonstrate the value of animated feedback—Vital makes visualization central to its workflow, while Pigments explicitly emphasizes reactive visual feedback—so FIELD can make the visualization itself the synthesis surface. [10]

Sound territory: liquid basses, moving growls, glassy geometries, organic drones, stereo shapes that move without chorus, fluid leads, alien formants, animated sub harmonics, and violently distorted terrain when trajectories cross high-gradient features.

Engineering specification

Item	Recommendation
DSP primitives	Phase accumulator; analytic scalar/vector field functions; Gaussian/RBF nodes; vortices; signed-distance primitives; procedural noise; path spline/circle/ellipse generators; interpolation; DC blocker
Anti-aliasing	Multi-resolution field/detail levels; select spatial LOD from scan velocity and note frequency; regularize singularities with ϵ; 2×–8× adaptive oversampling at high field gradients; final low-pass/downsample
Typical controls	**PATH** Linear/Circle/Orbit/Bezier; **SCAN** −100…+100%; **FORCE** 0…200%; **TEXTURE** 0.25×…8×; **FLOW** −4…+4 Hz; **SEPARATION** 0…100%
Prime modulation targets	Node X/Y, node force, path radius, path rotation, texture scale, flow velocity, turbulence, stereo separation
CPU	**Medium–High**, depending on analytic node count and oversampling
Determinism	Per-preset field seed + optional per-note variation
Stereo philosophy	Separate spatial pickups, **not** ordinary unison

A useful anti-alias rule is to think of FIELD's detail as spatial bandwidth. As pitch or trajectory velocity rises, very small terrain details must progressively disappear in the same way high harmonics must disappear near Nyquist. That philosophy is consistent with Vital's explicit Nyquist cutoff and with harmonic-domain approaches that generate only a finite set of legal partials. [11]

GENDRIFT — “A waveform that mutates every cycle but never loses the note.”

Tagline: Controlled mutation. Locked pitch.

GENDRIFT stores perhaps 3–64 control points inside exactly one oscillator period. Every completed period performs bounded stochastic mutations on their amplitudes and optionally their normalized time positions, producing a waveform that is always related to the previous cycle but is never merely an LFO scanning a static table.

This is directly inspired by the useful part of dynamic stochastic synthesis: SuperCollider's Gendy implementation describes control points whose amplitudes and durations can be modified from period to period, and explicitly notes a fixed-period variant in which points move within an unchanging cycle. [12] Origami should make **fixed pitch the default**, rather than allowing the stochastic timing system to make musical pitch wander.

Problems GENDRIFT solves: static or obviously periodic wavetable-position modulation; needing external LFO/random sources just to make an oscillator breathe; random systems that destroy pitch; note attacks changing unpredictably without a deterministic seed; large wavetable libraries merely to obtain subtle variation; expensive granular/spectral engines when the desired result is simply continuously non-repeating tone color. A Serum user specifically requested seeded, fixed randomization because re-randomization on every note could produce unwanted note-to-note variation; that is exactly the kind of control GENDRIFT should expose from day one. [13]

Sound territory: living analog-ish basses without analog emulation, gnarly irregular mid-basses, nervous leads, insect textures, slowly evolving pads, restrained “human” oscillator drift and completely unstable digital screams.

Item	Recommendation
DSP primitives	Fixed-period phase accumulator; control-point array; bounded random walk; selectable distributions; piecewise linear/cubic interpolation; deterministic PRNG
Anti-aliasing	Prefer cubic interpolation; correct hard corners where necessary; optional per-cycle FFT truncation to $N_{\max} < F_s/(2f_0)$; 2× quality mode for extreme mutation
Controls	**POINTS** 3–64; **AMP WALK** 0…100%; **TIME WALK** 0…100%; **CORRELATE** −100…+100%; **DISTRIBUTION** Linear/Cauchy/etc.; **SEED** 0…65535
Mod targets	Mutation depth, point count, selected point groups, correlation, skew/bias, distribution shape
CPU	**Low–Medium**
Killer feature	**FREEZE** instantly turns the current mutated state into a stable repeatable oscillator

SuperCollider warns that a high control-point count at high oscillator frequency can raise CPU cost substantially because points are being altered frequently, which argues for updating Origami's geometry once per completed cycle rather than running the stochastic model independently at every audio sample. [12]

QUASAR — “One pitch cycle becomes a microscopic sequence of explosions.”

Tagline: Pitch is the clock. Timbre is the event.

QUASAR creates a pitch-synchronous sequence of mathematical sound packets—windowed sinusoids, chirplets or damped resonant bursts—inside every fundamental period. The MIDI note determines **when a period repeats**, while a separate FORMANT/CARRIER parameter determines the internal packet frequency, giving users independent control over perceived pitch and spectral focus.

There is solid prior art for the underlying event concept: Csound's FOF generates successive sinusoidal bursts at a fundamental event rate while giving those bursts a separate formant center frequency; VOSIM likewise separates total event duration, which determines fundamental pitch, from individual pulse length, which establishes formant structure. [14] Origami's contribution would be a modern, bandlimited, aggressively modulatable **oscillator UX**, rather than presenting it as historical vocal synthesis or generic granular synthesis.

Problems QUASAR solves: difficulty separating bass pitch from formant identity; sample/grain dependence for vocal or creature-like sounds; high granular voice counts for something achievable analytically; static harmonic series; transient smearing from continuously morphed wavetables; requiring filters after the oscillator to construct moving resonant zones. Csound's FOF documentation explicitly shows fundamental event rate and formant center as independent controls, while its computation scales with overlapping burst density; this gives Origami a clear path to a controlled, compact implementation. [15]

Sound territory: talking bass, vowel growls, “laser throat” leads, metallic chirps, synthetic percussion, pitch-stable formant drones, nasal Reese layers and dubstep sounds where FORMANT sweeps dramatically while the actual MIDI pitch stays planted.

Item	Recommendation
DSP primitives	Pitch-period scheduler; analytic sine/chirp oscillator; window/envelope kernels; small overlap pool; polyphase packet renderer
Anti-aliasing	Reject/attenuate carrier trajectory above Nyquist; use bandlimited packet kernels; oversample chirps/highly asymmetric envelopes; harmonic-domain rendering for “Ultra” mode
Controls	**FORMANT** 0.25×…64× f_0 or 20 Hz…20 kHz; **BURSTS** 1–32; **DUTY** 1…100%; **DECAY** 0…100%; **CHIRP** −48…+48 st/event; **SKEW** −100…+100%
Mod targets	Formant, packet position, chirp, burst count, duty, stereo event offset, envelope curvature
CPU	**Low–Medium**

Historical VOSIM implementations are explicitly not fully bandlimited, so copying them literally would be a mistake; Origami should retain the event model but rebuild the packet renderer around strict Nyquist management. [16]

KINETIC — “Scan a machine while it moves.”

Tagline: Timbre has mass, tension and inertia.

KINETIC is an accessible reinterpretation of scanned synthesis. A network of masses connected by springs evolves relatively slowly, while the MIDI oscillator phase scans a path through the network at audio rate; therefore **the physical network determines timbre while scan rate determines pitch**.

Csound's official scanned-synthesis documentation uses exactly this useful separation: `scanu/scanu2` define and excite the mass/spring network while `scans` follows a trajectory through it. [17] This is substantially different from putting a resonator after a wavetable because the slowly evolving physical state itself becomes the waveform geometry.

Problems KINETIC solves: static oscillator timbres; modulation that has no physical memory/inertia; ordinary physical models in which pitch and body characteristics are too tightly coupled; lifeless pad motion requiring stacks of LFOs; lack of gesture-sensitive oscillator behavior; repetitive wavetable loops.

Sound territory: elastic bass, bowed-metal motion, huge breathing pads, plucked membranes that slowly deform, trembling sub layers, unstable industrial tones and expressive MPE patches where pressure actually “pushes” the virtual structure.

Item	Recommendation
DSP primitives	Mass/spring lattice; semi-implicit or Verlet integration; excitation impulse/noise; damping; closed scan paths; cubic interpolation
Anti-aliasing	Keep physical simulation substantially below audio rate; smooth scan interpolation; suppress lattice spatial modes that exceed the legal harmonic budget; optional 2× scan oversampling
Controls	**NODES** 8–128; **MASS** 0.1…10×; **TENSION** 0…100%; **DAMP** 0…100%; **EXCITE** 0…100%; **PATH** 0…100%
Mod targets	Local node force, tension, damping, excitation position, scan path, topology morph
CPU	**Medium**

AUTOMATA — “A rule generates the next waveform.”

Tagline: Every cycle is the next generation.

AUTOMATA represents one oscillator cycle as a row of discrete cells. At a selectable rate—every cycle, every fourth cycle, every 16 cycles, and so on—a cellular rule transforms that row into its next generation; the row is then interpreted through a **bandlimited reconstruction**, rather than played back naively as hard digital steps.

The result is deterministic evolution from an extremely small amount of state: a rule, seed and handful of macro parameters can generate long, structured transformations without a sample or wavetable file.

Problems AUTOMATA solves: presets that depend on external wavetable content; obvious “LFO up and down a table” movement; randomness with no long-term structure; expensive evolving textures; difficulty generating controlled glitch without sequencers; lack of repeatable generative behavior.

Sound territory: crawling digital basses, chiptune organisms, rhythmic harmonics, slowly crystallizing pads, deterministic glitches and patterns that sound “composed” despite originating inside a single oscillator.

Item	Recommendation
DSP primitives	1D cellular automaton/state grid; bit operations; seeded mutations; FFT or bandlimited basis reconstruction; phase accumulator
Anti-aliasing	Convert each generation to truncated Fourier coefficients or mipmapped periodic representations; crossfade only when a generation changes; avoid raw zero-order-hold cells
Controls	**RULE** 0…255 plus curated families; **CELLS** 16–256; **EVOLVE** 1/16…16 generations/cycle; **MUTATE** 0…100%; **SMOOTH** 0…100%; **SEED** 0…65535
Mod targets	Rule morph, generation rate, mutation, neighborhood bias, smoothness, seed offset
CPU	**Low**

HIVE — “Many voices behave like one organism.”

Tagline: Synchronization becomes timbre.

HIVE contains a population of perhaps 8–128 simple phase agents coupled to one another. Instead of ordinary unison, where independent detuned voices merely beat together, each agent's relative phase is influenced by its neighbors; positive coupling produces synchronizing clusters, negative or “frustrated” coupling keeps populations fighting one another.

The MIDI oscillator provides a common phase reference so the fundamental can remain musically stable while relative phase organization changes around it. Harmonic index, phase cluster, amplitude and stereo position can be assigned across the population, allowing timbre and width to emerge from **collective organization**, not chorus.

Problems HIVE solves: boring unison stacks; phase cancellation in wide basses; stereo width tied to detuning; static supersaw-like motion; CPU spent on full independent voices when each member only needs a small phase state; difficulty creating organic coordinated variation. A producer discussion on stereo synthesis specifically highlights the recurring challenge of obtaining very wide bass without destructive phase cancellation or obvious detuning. [18]

Sound territory: living Reese basses, swarms, murmuring pads, phase-organized supersaws, metallic clusters, unstable choruses and stereo sounds that can collapse to a deliberately designed mono core.

Item	Recommendation
DSP primitives	8–128 phase agents; coupled-phase update; harmonic-index table; SIMD sine bank; coherent stereo panner
Anti-aliasing	Never instantiate an agent harmonic above Nyquist; coupling operates on phase state rather than hard waveshaping; smooth agent births/deaths
Controls	**AGENTS** 4–128; **COUPLE** −200…+200%; **DISORDER** 0…100%; **CLUSTERS** 1–8; **FRICTION** 0…100%; **WIDTH** 0…100%
Mod targets	Coupling, disorder, cluster count, population distribution, harmonic spread, stereo radius
CPU	**Medium–High**, highly SIMD-friendly

For implementation culture, Vital is a useful benchmark: its official feature page specifically advertises SSE optimization to keep oscillator unison inexpensive. HIVE should similarly be conceived as a vectorized oscillator bank from the beginning rather than as 128 generic voice objects. [19]

RECURSION — “The next cycle remembers the last.”

Tagline: Sound with heredity.

RECURSION's state is the **previous oscillator period**. At every wrap, Origami applies a bounded transformation—rotate, contract, diffuse, fold, inject, differentiate, blur, phase-scramble within limits—and uses the result as the next cycle, meaning the oscillator has genuine causal memory.

A normal wavetable animation asks, “which stored frame comes next?” RECURSION asks, **“what does this cycle become because of what it was?”** The difference becomes especially obvious under modulation: a momentary gesture can permanently alter the oscillator state until it decays, freezes or resets.

Problems RECURSION solves: motion that immediately returns when an LFO returns; inability for modulation gestures to leave a sonic trace; huge animated wavetable banks; static note sustains; lack of oscillator-scale feedback processes; resampling workflows used merely to “commit” transformations. A Serum 2 user reported that FM resampling could differ from the live sound and that unsynchronized modulation could drift, illustrating why a deterministic stateful oscillator could be useful without a separate resample-and-reimport cycle. [20]

Sound territory: self-evolving basses, continuously decaying complexity, recursive metallic tones, patches that “remember” aftertouch, fractal-ish transitions and one-knob descent from smooth tone into destruction.

Item	Recommendation
DSP primitives	Per-voice cycle buffer; FFT/IFFT or time-domain transform bank; feedback state; deterministic injection source; normalization
Anti-aliasing	Truncate transform spectrum every generation to note-dependent N_\max; oversample nonlinear transforms; enforce seamless cycle boundary; DC/high-frequency energy guards
Controls	**MEMORY** 0…99.5%; **FOLD** 0…200%; **ROTATE** −100…+100% cycle; **DIFFUSE** 0…100%; **INJECT** 0…100%; **RESET** Note/Beat/Free
Mod targets	Every transformation component plus freeze/reset trigger
CPU	**Medium**, rising to Medium–High for FFT-heavy modes

Finite Fourier representations provide a clean safety mechanism here: recent work demonstrates arbitrary periodic waveforms represented and transformed using only legal Fourier coefficients, avoiding components above Nyquist at the representation stage. [21]

CIPHER — “Sound generated by a rule, not a wave.”

Tagline: Compute the oscillator.

CIPHER takes a pitch-synchronous integer phase counter and feeds it through a tiny visual Boolean/bitwise state machine—XOR, shift, carry, mask, rotate, compare, divide—then maps that state to amplitude or a small bandlimited basis. Crucially, users do **not** type bytebeat equations; Origami exposes curated musical macros while showing the logic network visually.

Problems CIPHER solves: glitch synthesis requiring external sequencers or effects; naive digital/bit oscillators that alias aggressively; formula-based procedural synthesis that is inaccessible to non-programmers; static wavetable loops; CPU-heavy methods for simple evolving digital patterns; lack of deterministic rhythmic timbre at oscillator level.

Sound territory: machine basses, modem shrieks, computational growls, bit-crushed leads without a bitcrusher, deterministic rhythmic harmonics and “broken computer” textures that still follow MIDI perfectly.

Item	Recommendation
DSP primitives	Integer phase counter; bitwise instruction graph; small state registers; transition detector; bandlimited output renderer
Anti-aliasing	Treat state transitions as bandlimited events or render one logical period into truncated Fourier coefficients; never simply output arbitrary hard integer steps at audio rate
Controls	**RULE** curated 0…255 set; **BITS** 2–24; **SHIFT** −16…+16; **DIVIDE** 1…64; **CARRY** 0…100%; **SMOOTH** 0…100%

Item	Recommendation
Mod targets	Rule morph, bit depth, operator mix, divider, state feedback, seed
CPU	**Low**

COLLIDER — "Impacts are the waveform."

Tagline: Build a chamber. Fire the particles.

COLLIDER contains phase-normalized particles traveling through a small two-dimensional chamber. Crossing boundaries, colliding with obstacles or entering force zones emits **bandlimited impulses or resonant packets**; the arrangement and timing of those interactions constitute the waveform.

Unlike FIELD, whose signal is a continuous sample of spatial energy, COLLIDER is an **event system**: geometry determines when events occur. Unlike granular synthesis, there is no source audio being cut into grains—the events are generated mathematically at collision times.

Problems COLLIDER solves: poor oscillator-level transient design; needing samples for clicky/percussive timbres; lack of intuitive control over sparse versus dense spectra; sterile repeated transients; limited physical interaction without a full physical model; difficulty producing rhythmic microstructure independent of the host sequencer.

Sound territory: metal ticks, laser impacts, hard neuro-bass transients, synthetic percussion, rattling chambers, crackling leads and sounds somewhere between oscillator, percussion generator and microscopic pinball machine.

Item	Recommendation
DSP primitives	Small particle integrator; analytic geometry/SDF collision detection; event scheduler; bandlimited impulse; damped sine kernels
Anti-aliasing	Emit pre-bandlimited impulses rather than one-sample deltas; clamp ring frequency below Nyquist; oversample only nonlinear collision response
Controls	**PARTICLES** 1–64; **SPEED** 0.25×…8×; **GRAVITY** −200…+200%; **BOUNCE** 0…100%; **RING** 0…100%; **CHAMBER** geometry selector
Mod targets	Obstacle locations, particle velocity, force zones, damping, collision resonance, stereo impact position
CPU	**Medium**

DAEMON — "A tiny nonlinear machine that self-oscillates."

Tagline: Build instability. Force it to sing in tune.

DAEMON is the highest-risk concept: each voice contains a small nonlinear state-space network made from integrators, delays, saturating junctions, comparators and feedback paths. Instead of emulating a Minimoog-style oscillator, it intentionally creates unfamiliar self-oscillating dynamical systems and then uses time normalization or a pitch servo to keep their dominant cycle tied to MIDI.

This should feel almost alive: a parameter can change not only harmonic content but the **behavioral regime** of the machine—stable orbit, period doubling, quasi-periodic state or noise-like instability. That places it well outside ordinary wavetable warping, although its higher engineering risk makes it inappropriate as the first Origami prototype.

Problems DAEMON solves: "digital and crisp" source character requiring downstream saturation; no oscillator memory/hysteresis; static periodic behavior; predictable warp curves; lack of near-chaotic transitions; reliance on effects to create nonlinear personality. One long-running KVR discussion includes users describing a struggle to get less bright/crisp wavetable tones and recommending filtering/saturation as additional processing; DAEMON attacks that requirement at the generator itself instead of bolting coloration on afterward. [22]

Sound territory: unstable analog-adjacent tones that are not analog emulations, tearing feedback basses, squealing leads, subharmonic states, chaotic transitions, electrical drones and presets with dramatic "edge of collapse" performance macros.

Item	Recommendation
DSP primitives	Small state-space graph; integrators; nonlinear junctions; bounded feedback delay; zero-crossing estimator; pitch servo/time-scale normalization
Anti-aliasing	4×–16× adaptive oversampling; bandlimited discontinuities; carefully chosen smooth nonlinearities where possible; steep decimation filter
Controls	**TOPOLOGY** 1–16; **DRIVE** 0…24 dB; **FEEDBACK** 0…99.5%; **BIAS** −100…+100%; **DAMP** 0…100%; **LOCK** 0…100%
Mod targets	Every network coefficient, feedback path, topology morph, servo strength, nonlinear threshold
CPU	**High**

Shared DSP architecture and prototype signal flows

The most important engineering decision is **not** to build ten entirely independent oscillator codebases. Origami should supply a common oscillator substrate that each mode plugs into:

MIDI pitch → high-precision phase clock → mode state/model → mode renderer → anti-alias stage → native stereo mapping → common level/pan/unison/output

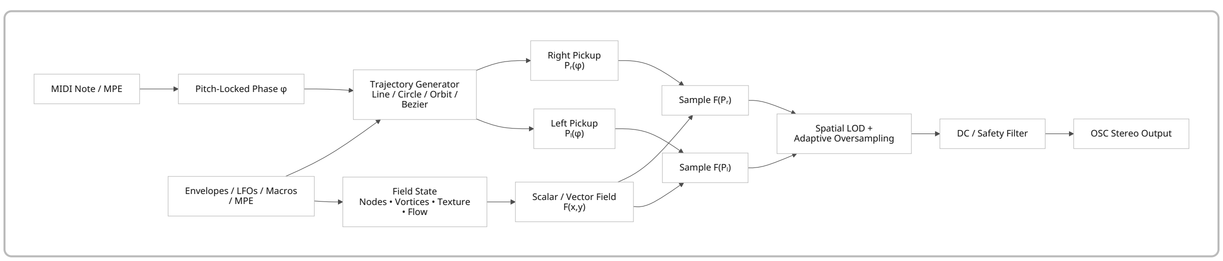
That common kernel should provide deterministic seeded randomness, host-sample-rate independence, retrigger modes, MPE inputs, a note-dependent harmonic budget, oversampling services, test instrumentation and a standardized modulation interface. This makes it much easier for unusual engines to remain “synth-like” even when their mathematical representation is alien.

The quality philosophy should be **selective rather than brute force**. Community discussions show the tension clearly: users notice oscillator aliasing, but some explicitly object to forced oversampling because of CPU cost. [23] Phase Plant therefore offers explicit wavetable bandlimiting before heavy phase modulation, while Vital advertises a sharp Nyquist cutoff alongside CPU-conscious oscillator implementation. [24] Even though Origami's design brief imposes no hard CPU constraint, selective antialiasing will still matter when users combine high polyphony, unison and many oscillator modules.

A practical common **quality stack** is:

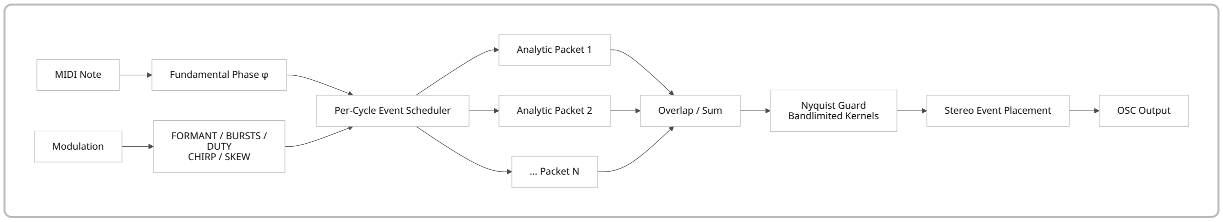
1. **Prevent illegal energy when possible.** In harmonic or reconstructed modes, generate only components below the note-specific Nyquist budget; additive research demonstrates the strength of representing arbitrary periodic waveforms using finite Fourier coefficients. [21]
2. **Bandlimit state representations.** FIELD uses spatial LOD; AUTOMATA uses Fourier reconstruction; RECURSION truncates each cycle's spectrum.
3. **Correct sparse discontinuities rather than oversampling everything.** CIPHER/COLLIDER can use bandlimited event kernels at transition/impact times.
4. **Oversample genuinely nonlinear systems.** DAEMON, hard FIELD singularities and aggressive RECURSION operations deserve 4×–16× paths.
5. **Give the user Eco / Normal / Ultra**, but make Normal musically clean enough that “turn on 8× oversampling” is not normal workflow. Community concern about forced oversampling and Xfer's reports of heavy CPU use from complex voices both favor adaptive quality rather than an all-or-nothing approach. [25]

FIELD signal flow



The crucial engineering contract is that **field evolution must never own pitch**. The environment may flow slowly or behave chaotically, but ϕ remains the authoritative musical clock. That is what prevents FIELD from degenerating into a visually impressive noise generator.

QUASAR signal flow



This second architecture is attractive because **fundamental and formant are structurally independent** rather than being separated later with filtering. FOF and VOSIM provide established examples of pitch-period events whose internal sinusoidal/pulse frequency controls spectral/formant placement independently of fundamental event timing. [14]

Community pain points and product implications

These are **reported community experiences, not universal measurements**. Their value is that they expose recurring friction points around oscillator design that Origami can deliberately address.

Reported pain point	Community evidence	Origami design response
“Another wavetable synth” does not automatically justify itself.	In an r/synthesizers Pigments launch discussion, users said they could already achieve much of what they heard elsewhere, described a glut of wavetable synths, and said a newcomer needed to do “something extra special.” [26]	Make FIELD/GENDRIFT/etc. visibly different synthesis objects, not a larger warp menu.
Users themselves question what wavetable synthesis uniquely accomplishes.	A 2025 KVR discussion begins by asking what sounds wavetable can make that other methods cannot; one experienced poster argues much wavetable usage is reducible to other synthesis/filter approaches and specifically points to warping/FM alias issues. [27]	A top-level Origami mode must change the *representation* of sound, not just offer a shortcut to another spectral shape.
Aliasing from aggressive oscillator manipulation is audible enough to become forum discussion.	KVR users discussed wavetable aliasing directly; separately, Phase Plant's official documentation includes a sharp pre-phase-modulation bandlimit option, confirming that heavy phase modulation creates a practical bandlimiting concern. [28]	Publish a per-mode antialias contract and build note-aware quality management into the oscillator kernel.

Reported pain point	Community evidence	Origami design response
Brute-force oversampling has a CPU/workflow cost.	One KVR participant explicitly objected to forced oversampling because of CPU impact. [23]	Use structural antialiasing first; make oversampling adaptive and user-selectable rather than mandatory everywhere.
Complex oscillator modes can explode CPU through polyphony × unison × internal voices.	Xfer staff says high unison/grain counts plus long releases can produce very high active voice counts, and specifically calls Serum 2's Granular and Spectral oscillator types relatively demanding. [29]	Track **effective internal element count**, not just MIDI polyphony; vectorize HIVE, cap inactive particles, and sleep stable state simulations.
Random phase behavior can make otherwise identical notes sound inconsistently different.	An Xfer user requested fixed seeded random oscillator phases because per-note re-randomization sometimes produced unwanted variation. [13]	Every stochastic Origami mode gets **Seed: Fixed / Per Note / Free**, plus a visible FREEZE function.
Modulated/resampled oscillator states can be hard to reproduce exactly.	A Serum 2 forum user reported FM resamples differing from the live patch and low-frequency modulation drifting relative to the resampling window. [20]	Make cycle-state modes explicitly phase-locked and export/freeze the actual internal state, not an arbitrary time slice.
Wide bass without destructive mono behavior remains a practical concern.	A drum-and-bass producer described wide bass without phase cancellation/ detuning as a long-standing problem in a Reddit discussion. [18]	FIELD uses spatially separated pickups; HIVE controls phase coherence; both should expose a correlation/mono meter inside the oscillator.
Some users perceive wavetable sources as bright/crisp and reach for post-processing to soften them.	A KVR user described struggling to make Serum warm because patches sounded bright/crisp; replies recommended filtering and saturation. [22]	KINETIC/GENDRIFT/DAEMON can generate complexity with natural temporal/state structure rather than needing upper-harmonic-heavy static sources followed by coloration.
Wavetable content workflows themselves can create friction.	An Xfer forum thread documents a single-cycle export/reimport problem ultimately tied to wavetable metadata and import handling. [30]	Several new modes—FIELD, AUTOMATA, CIPHER, GENDRIFT—should be completely procedural and require no external table format at all.

There is also a UI lesson hidden in the community feedback. In the Pigments discussion, even users who viewed the underlying synthesis as similar to Serum highlighted modulation visualization and routing as meaningful differentiation. [26] Arturia itself now markets Pigments around a reactive, color-coded interface in which animation helps clarify sound design, while Vital similarly makes animated controls, waveforms and spectrograms part of the core product identity. [31] For Origami, that means **FIELD's moving landscape, GENDRIFT's evolving polygon, KINETIC's moving lattice and AUTOMATA's generations should be the controls**, not merely pretty scopes sitting above conventional knobs.

Naming system and prototype recommendation

Overall family-label candidates

For the small UI label that answers "what kind of oscillator is this?", these are the strongest options:

Candidate	Character	Verdict
FORM	Very Origami-specific; concise	**Best branding choice**
ENGINE	Immediately understandable	Best conventional choice
DOMAIN	Communicates fundamentally different representations	Strong technical/modern option
CORE	Compact, aggressive	Good
SYSTEM	Especially suitable for FIELD/KINETIC/HIVE	Good, slightly clinical
MODEL	Accurate for generative architectures	Clear but less exciting
GENERATOR	Conventional audio terminology	Probably too long
MATTER	Strange and brandable	Interesting, less immediately clear

My preference is **FORM**. It connects directly to "Origami" without forcing gimmicky terminology:

```
FORM  WAVETABLE
FORM  GRANULAR
FORM  SPECTRAL
FORM  FIELD
FORM  GENDRIFT
```

It also subtly reinforces the design philosophy: these are different **forms of generating sound**, not effect algorithms.

Alternate individual-mode name pool

The selected names are deliberately short enough to fit the same oscillator header structure. For the concepts most likely to change during branding, these are the strongest alternates:

Selected	Strong alternates
FIELD	DOMAIN · FLUX · TERRAIN
GENDRIFT	MUTANT · GENESIS · DRIFT
QUASAR	PULSAR · BURST · PULSEFORM
KINETIC	TENSOR · LATTICE · SCANMASS
HIVE	SWARM · COLONY · SYNCHRON
AUTOMATA	CELL · MACHINE · RULE
RECURSION	INHERIT · ECHOFORM · MEMORY
DAEMON	BLACKBOX · ABYSS · CIRCUIT

If the goal is a cohesive intimidating family, my favorite final set is:

WAVETABLE · GRANULAR · SPECTRAL · FIELD · GENDRIFT · QUASAR · KINETIC · AUTOMATA · HIVE · RECURSION · COLLIDER · CIPHER · DAEMON

They sound like **things**, rather than DSP functions. That is desirable: `SPECTRAL SHIFT` sounds like a processor; `HIVE` sounds like an instrument behavior.

The three prototypes I would actually build first

FIELD — flagship differentiation. This is the concept most capable of becoming visually synonymous with Origami. Competitors already compete heavily on visual synthesis workflow, but the mainstream products surveyed still center their oscillator taxonomies around waveforms, samples, grains, harmonics and familiar physical models. [1] FIELD gives Origami a screenshot-able and explainable proposition: **"sound is a space you move through."** Start only with a scalar field, six node primitives and Linear/Circle paths; vector flow can be version two.

GENDRIFT — lowest-risk proof that Origami means genuinely different oscillation. The technical precedent is mature enough that the hard conceptual questions are understood: evolving control-point geometry works, fixed-period behavior is possible, and CPU increases predictably with point count/ frequency. [12] Origami can make it dramatically more musical by guaranteeing MIDI pitch, adding deterministic seeds and applying modern antialiasing. It should be possible to know within the first few weeks whether this engine has a distinct sonic identity.

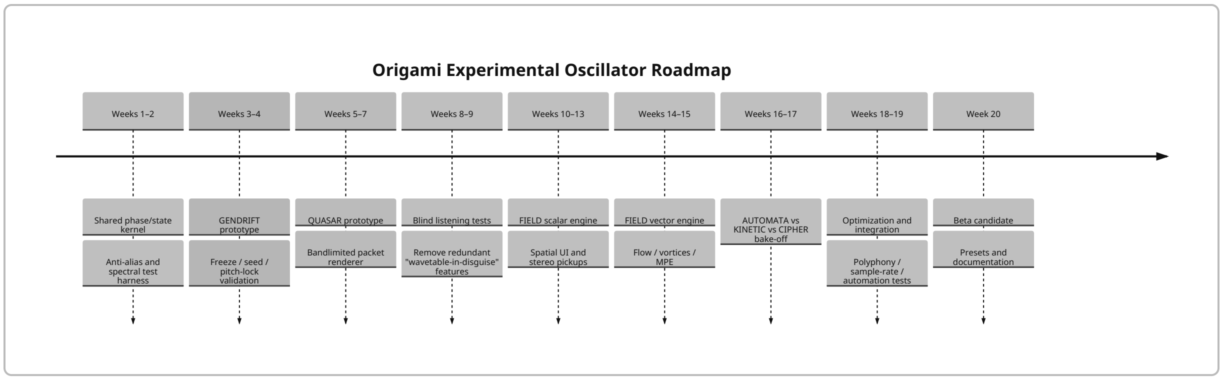
QUASAR — highest immediate sound-designer payoff. Unlike FIELD, which requires substantial experimentation before presets become obvious, QUASAR has a predictable sweet spot: pitch-independent formant motion, aggressive bass/vocal timbre and sharply articulated transients. FOF and VOSIM demonstrate that the underlying event/formant relationship is computationally tractable and musically productive; Origami's job is to make it modern, bandlimited and performance-oriented. [14]

I would **not** prototype DAEMON first despite its novelty. Its pitch servo, nonlinear stability, modulation smoothing and anti-aliasing requirements could consume months before answering the fundamental product question, “does this make better sounds?” Likewise, I would hold HIVE until Origami's stereo/mono framework is mature, because its main advantage depends on unusually careful phase and stereo behavior.

Implementation roadmap

The following is a realistic **rough 20-week R&D sequence** assuming one engineer focused primarily on DSP and another person able to work on oscillator UI/visualization and integration in parallel. It is a planning estimate rather than a benchmark; a solo implementation would naturally stretch longer.

Window	Milestone	Engineering exit criteria
Weeks 1–2	Common oscillator kernel	64-bit phase accumulator; deterministic PRNG; per-note state; retrigger/free modes; common oversampler; Nyquist-budget API; modulation smoothing; automated spectral test harness
Weeks 3–4	GENDRIFT proof	Fixed pitch from sub to upper keyboard; 3–64 evolving points; deterministic seed; Freeze; no obvious aliasing in Normal mode; preset-worthy bass/pad/lead examples
Weeks 5–7	QUASAR proof	Independent pitch/formant control; analytic packet pool; chirp; stable automation; packet Nyquist protection; mono/stereo versions
Weeks 8–9	Listening and kill gate	Compare modes blind against Wavetable/Spectral/Granular equivalents; remove controls that merely reproduce existing warps; define 20 factory “identity” patches per successful mode
Weeks 10–13	FIELD scalar prototype	Analytic nodes; Linear/Circle/Bezier scan; spatial LOD; visual white→red energy display; dual stereo trajectories; click/drag field editing
Weeks 14–15	FIELD vector extension	Vortices, attractors/repellers, FLOW and trajectory bending; high-gradient adaptive quality; MPE spatial modulation
Weeks 16–17	Second-wave bake-off	Minimal AUTOMATA, KINETIC and CIPHER implementations; spend ≤1 week each before choosing one for production
Weeks 18–19	Production hardening	Polyphony/unison profiling; denormal handling; sample-rate changes; preset migration; automation torture tests; mono compatibility; silence/state sleeping
Week 20	Beta candidate	Three polished new modes, factory presets, documentation, CPU quality modes, deterministic recall and unified Origami visual language



A particularly important milestone is the **Weeks 8–9 kill gate**. The user concern that began this research—"couldn't tension just be spectral shift?"—should become a formal product test. For every candidate, create the closest equivalent patch possible in Origami's existing Wavetable/Spectral/Granular modes. If users cannot reliably hear or understand why the new engine behaves differently, **demote its interesting functions into processors and kill the oscillator mode**. That standard is stricter than adding features for a spec sheet, but community discussions around wavetable synth saturation strongly suggest that differentiation has to be perceptible rather than nominal. [32]

The corresponding technical acceptance test should be equally unforgiving: sweep every oscillator from roughly MIDI C0 through the top of the practical keyboard at 44.1, 48, 88.2, 96 and 192 kHz; automate every continuous parameter at audio and control rates where supported; exercise maximum polyphony; test fixed and randomized seeds; null-test deterministic retriggers; inspect energy above the legal harmonic budget; verify mono collapse; and profile Eco/Normal/Ultra independently. The emphasis on alias-controlled oscillator generation in Vital, explicit bandlimiting in Phase Plant, and current work on alias-free arbitrary periodic oscillation provides a strong benchmark: exotic synthesis does not need to mean dirty implementation unless dirt is deliberately requested. [2]

The resulting product story is much stronger than "Origami has more oscillator modes." It becomes:

> **Wavetable manipulates shape. Granular manipulates time. Spectral manipulates frequency. FIELD manipulates space. GENDRIFT manipulates evolution. QUASAR manipulates events. KINETIC manipulates matter. AUTOMATA manipulates rules. HIVE manipulates populations. RECURSION manipulates memory.**

That is the architecture I would use as the north star for Origami: **a synthesizer where changing oscillator type genuinely changes what the user believes sound is made of.**

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